Title

Early Detection of Emerging Misinformation through Narrative Evolution and Semantic Drift: A Temporal Graph-Based Approach

2. Exploratory Data Analysis: PHEME Dataset

2.1 Source and Composition

The PHEME Rumour and Non-Rumour (RNR) dataset consists of Twitter conversation threads collected during five real-world breaking-news events: Charlie Hebdo, Ferguson, Germanwings Crash, Ottawa Shooting, and Sydney Siege. Each discussion thread is rooted in a source tweet that is annotated as either a rumour or a non-rumour. All subsequent replies within the thread inherit the label assigned to the source tweet. The original corpus contains 103,212 tweets distributed across these five events.

The dataset is particularly suitable for misinformation research because it preserves the conversational structure of online discussions, enabling analysis of both content and diffusion dynamics. After preprocessing, the final dataset contains 94,395 English-language tweets organized into 5,802 discussion threads involving 45,154 unique users.

Table 1

Final Dataset Summary

Metric	Value
Tweets	94,395
Threads	5,802
Events	5
Users	45,154
Rumour Tweets	28,680
Non-Rumour Tweets	65,715

2.2 Data Preparation

Several preprocessing steps were applied to ensure data quality and consistency. First, only English-language tweets were retained, reducing the dataset from 103,212 to 95,167 tweets.

Tweet timestamps were then converted to timezone-aware datetime objects, with no invalid timestamps identified.

All identifier columns were converted to string format to preserve the integrity of large Twitter snowflake IDs. Duplicate tweets were subsequently removed. Investigation revealed that 772 tweets appeared twice due to a known characteristic of the PHEME dataset in which source tweets are duplicated within the reactions folder of their own thread. In such cases, the source-tweet version was retained and the duplicated reaction entry was removed. This produced a final dataset containing 94,395 tweets.

Tweets were then sorted chronologically within each thread, and each observation was linked to its corresponding source-tweet timestamp. This enabled the computation of temporal features such as time elapsed since thread initiation.

2.3 Class Distribution and Dataset Structure

The final dataset contains 28,680 tweets belonging to rumour threads and 65,715 tweets belonging to non-rumour threads. At the thread level, 1,972 threads are labelled as rumours and 3,830 as non-rumours, corresponding to a rumour proportion of approximately 34%.

Table 2

Thread Level Distribution

Level	Rumours	Non-rumours	Rumour share
Tweets	28,680	65,715	30.4%
Threads	1,972	3,830	34.0%

Thread-level distributions are more meaningful than tweet-level distributions because labels are assigned to discussion threads rather than individual tweets. Analysis of thread sizes indicates that non-rumour discussions are generally larger, with a median of 13 tweets per thread compared with 11 tweets for rumour discussions. However, both classes exhibit strongly right-skewed distributions, with a small number of highly active conversations generating substantially larger cascades.

The dataset contains 5,802 source tweets and 88,593 replies. A total of 401 threads (6.91%) consist solely of the source tweet and contain no replies. Author participation is broadly similar across classes, with both rumour and non-rumour discussions involving a median of approximately ten unique users per thread.

Event-level analysis reveals notable variation across the five news events. Charlie Hebdo and Ferguson are predominantly composed of non-rumour discussions, whereas Germanwings Crash and Ottawa Shooting contain larger proportions of rumour content. Sydney Siege remains non-rumour dominant but exhibits a relatively higher rumour presence than Charlie Hebdo and Ferguson. These differences suggest that misinformation prevalence varies across events and highlight the importance of considering event context during model development and evaluation.

Table 3

Event × Label Distribution (English-Only Processed Dataset)

Event	Non-rumours	Rumours
Charlie Hebdo	27,895	6,543
Ferguson	16,388	6,077
Germanwings Crash	1,685	2,222
Ottawa Shooting	5,425	5,920
Sydney Siege	14,322	7,918

2.4 Temporal Characteristics

Temporal analysis reveals substantial differences between rumour and non-rumour discussions. Non-rumour threads exhibit a median duration of 3.52 hours, whereas rumour threads exhibit a median duration of 1.67 hours. Although some discussions remain active for several weeks, most conversations conclude within a relatively short period after the source tweet is posted.

Table 4

Median temporal characteristics of rumour and non-rumour discussion threads

Metric	Non-rumours (Median)	Rumours (Median)
Thread Duration (hours)	3.52	1.67
Time to First Reply (minutes)	2.60	1.75
Inter-Reply Gap (minutes)	1.95	1.60
Thread Half-Life (minutes)	25.73	13.02

Rumour threads also attract engagement more rapidly. The median time to first reply is 1.75 minutes for rumours compared with 2.60 minutes for non-rumours. Similarly, inter-reply intervals are shorter for rumour discussions, indicating more concentrated bursts of activity.

To further examine diffusion speed, thread half-life was calculated as the time required for a discussion to accumulate 50% of its eventual replies. Rumour threads exhibit a median half-life of 13.02 minutes, while non-rumour threads require 25.73 minutes. This nearly two-fold difference suggests that rumours spread and attract attention considerably faster than non-rumours during the early stages of discussion.

Early engagement analysis further supports this observation. Approximately 67% of all threads receive at least one reply within five minutes of the source tweet, increasing to more than 89% within one hour.

Table 5

Early reply-growth coverage across discussion threads

Time Since Source Tweet (minutes)	**5**	**15**	**30**	**60**
Threads with ≥1 Reply (%)	67.1	81.8	86.5	89.1

These findings indicate that a substantial proportion of conversational activity occurs very early in a thread's lifecycle, making early detection approaches both practical and potentially effective.

2.5 Structural Characteristics

To analyse conversational structure, each discussion thread was represented as a directed reply graph in which nodes correspond to tweets and edges represent reply relationships. Several structural properties were computed for each thread, including maximum depth, maximum width, and branching factor.

The median values of these measures are broadly similar across classes. Both rumour and non-rumour discussions exhibit a median depth of two reply levels, while median widths are eight and seven respectively. Median branching factors are also nearly identical.

Although aggregate structural measures provide limited class separation, differences emerge when examining cascade composition. Approximately 58.47% of replies within rumour discussions are directed to the source tweet, compared with 50.93% for non-rumours.

Table 6

Structural Characteristics of Discussion Threads

Metric	Non-rumours	Rumours
Median Maximum Depth	2	2
Median Maximum Width	8	7
Median Branching Factor	3	3
Direct Replies to Source (%)	50.93	58.47
Reply-to-Reply Interactions (%)	49.07	41.53

Conversely, non-rumour discussions contain a larger proportion of reply-to-reply interactions. This suggests that rumour cascades tend to exhibit a fan-out structure centred on the original claim, whereas non-rumour discussions are more likely to develop into extended conversational exchanges.

2.6 Semantic Characteristics

Semantic analysis indicates relatively small differences between the two classes at the individual tweet level. Non-rumour tweets contain an average of 14.9 words per tweet, compared with 14.6 words for rumours. Median tweet lengths are identical at approximately fifteen words, indicating that the central tendency of tweet length is effectively the same across both classes.

Entity usage patterns are also similar. Hashtag frequency is nearly identical across classes, while mentions and media usage differ only marginally. The most notable distinction is a slightly higher prevalence of URLs within rumour discussions.

Table 7

Semantic characteristics of rumour and non-rumour discussions

Metric	Non-rumours	Rumours
Mean Words per Tweet	14.9	14.6
Median TTR	0.600	0.594
Mean URLs per Tweet	0.09	0.13
Mean Mentions per Tweet	1.73	1.62

Frequently occurring hashtags are largely event-specific and reflect the underlying news stories represented in the dataset, including hashtags such as #ferguson, #charliehebdo, and #sydneysiege. Rumour discussions additionally contain a greater presence of generic breaking-news hashtags.

Vocabulary diversity was assessed using the type-token ratio (TTR) at the thread level. Median TTR values are nearly identical for rumours (0.594) and non-rumours (0.600), indicating comparable levels of lexical diversity. Overall, semantic characteristics alone provide limited separation between the two classes.

2.7 User Characteristics

User-level features were examined to determine whether participant characteristics differ across rumour and non-rumour discussions. Median follower counts are similar across classes, with values of 461 and 423 for non-rumours and rumours, respectively. Comparable patterns are observed for friend counts and account age.

Table 8

User Characteristics of Discussion Participants

Metric	Non-rumours	Rumours
Median Followers	461	423
Median Friends	484	472
Verified Accounts (%)	4.23	5.91
Median Account Age (days)	1,244	1,282

The proportion of verified accounts is slightly higher within rumour discussions (5.91%) than non-rumour discussions (4.23%). This may reflect the involvement of news organizations, public figures, and other high-visibility accounts during the dissemination of breaking information.

Although median follower counts are comparable, mean follower counts differ substantially between the two classes. This discrepancy is driven by a small number of highly influential accounts with exceptionally large audiences, illustrating the highly skewed nature of social media influence.

2.8 Summary of Findings

The exploratory analysis identifies temporal behaviour and conversational structure as the most informative distinctions between rumour and non-rumour discussions. Rumour threads attract replies more quickly, exhibit shorter inter-reply intervals, accumulate engagement at a faster rate, and reach half of their final activity considerably sooner than non-rumour threads. Structurally, rumour discussions contain a greater proportion of direct responses to the source tweet, whereas non-rumour discussions display more reply-to-reply interactions.

In contrast, semantic characteristics such as tweet length, lexical diversity, and entity usage exhibit only minor differences between classes. User-level attributes are similarly comparable, with the exception of a slightly higher proportion of verified accounts participating in rumour discussions.

These findings suggest that temporal evolution and discussion structure may provide stronger early indicators of misinformation than static content or user-profile features. Consequently, subsequent stages of this research focus on modelling the evolution of conversational dynamics, semantic change, and narrativity within online discussion threads.

3. Exploratory Data Analysis: NarraDetect Dataset

3.1 Source and Composition

The NarraDetect dataset was used to model narrativity, defined as the extent to which a text exhibits narrative characteristics such as agency, event sequencing, and world-building. The dataset consists of two complementary files. The first, NarraDetect Large, contains 13,543 text passages across 18 genres with binary narrative (POS) and non-narrative (NEG) labels. The second, NarraDetect Scalar, contains 394 passages independently scored by three human annotators on multiple narrativity dimensions using a five-point Likert scale.

The two files are disjoint and serve different purposes. NarraDetect Large provides a large corpus suitable for supervised learning, while NarraDetect Scalar provides human-grounded narrativity scores that can be used to validate the construct measured by the dataset.

Table 9

NarraDetect Dataset Summary

Dataset	Rows	Genres	Unit of Analysis
NarraDetect Large	13,543	18	Text Passage
NarraDetect Scalar	394	18	Text Passage

3.2 Data Preparation

Several preprocessing steps were applied prior to analysis. Column names were standardised, duplicate identifiers were checked, and both datasets were converted to a consistent schema. No duplicate IDs were identified, and there was no overlap between passages contained in the Large and Scalar datasets. Five-fold stratified splits were also created to support future model evaluation while preserving genre distributions.

3.3 Label Distribution

The NarraDetect Large dataset contains 8,373 narrative passages and 5,170 non-narrative passages, corresponding to a narrative proportion of 61.83%. The Scalar dataset exhibits a more balanced distribution, with approximately 54% of passages classified as narrative according to human-derived labels.

Table 10

Label Distribution

Dataset	Narrative (POS)	Non-Narrative (NEG)	Narrative Share
Large	8,373	5,170	61.83%
Scalar (Human Labels)	214	180	54.31%

The relatively balanced class distributions reduce concerns regarding severe class imbalance and provide suitable conditions for training and evaluating narrativity classifiers.

3.4 Genre–Label Relationship

A notable characteristic of NarraDetect Large is that labels are entirely determined by genre. Genres such as novels, fairy tales, historical narratives, Reddit stories, and story-cloze tasks are consistently labelled as narrative, whereas legal documents, court opinions, philosophical texts, and academic abstracts are labelled as non-narrative.

This observation indicates that the Large dataset represents a genre-based approximation of narrativity rather than direct human assessment. Consequently, the Scalar dataset provides an important complementary source of human-grounded narrativity measurements.

3.5 Text Length Characteristics

NarraDetect passages are substantially longer than tweets contained in the PHEME dataset. The median passage length is 104 words in NarraDetect Large and 111 words in NarraDetect Scalar, compared with approximately 15 words per tweet in PHEME.

Table 11

Text Length Comparison

Dataset	Median Words	Mean Words
NarraDetect Large	104	115.6
NarraDetect Scalar	111	121.9
PHEME Tweets	15	~15

Within NarraDetect, narrative passages are generally shorter than non-narrative passages. This pattern is primarily driven by genre differences, as legal and academic texts tend to be longer and are predominantly classified as non-narrative.

3.6 Narrativity Scores

The Scalar dataset provides continuous narrativity scores through the variable avg_overall, which represents the average assessment of three human annotators. The distribution of these scores exhibits clear separation between narrative and non-narrative passages, with narrative texts receiving substantially higher ratings across all narrativity dimensions.

Table 12

Average Narrativity Scores by Human Label

Human Label	Agency	Event Sequencing	World-Making	Overall
Non-Narrative	1.81	2.15	1.73	1.90
Narrative	4.29	4.30	4.35	4.31

The large separation between the two groups suggests that the dataset successfully captures meaningful variation in narrativity and provides a strong foundation for supervised modelling.

3.7 Summary of Findings

The exploratory analysis demonstrates that NarraDetect provides a suitable foundation for modelling narrativity. The dataset contains a large collection of labelled passages spanning multiple genres, exhibits relatively balanced class distributions, and includes human-grounded narrativity scores that strongly differentiate narrative and non-narrative texts. The Scalar dataset further confirms that agency, event sequencing, and world-making collectively capture a coherent narrativity construct.

For the purposes of this research, NarraDetect serves as the source dataset for learning narrative characteristics that can subsequently be transferred to PHEME discussion threads. The resulting narrativity estimates will later be incorporated into the analysis of misinformation diffusion, semantic evolution, and conversational dynamics.

4. Cross-Dataset Observations

The exploratory analysis highlights the complementary roles of the PHEME and NarraDetect datasets within the proposed research framework.

PHEME provides information about the temporal and structural evolution of online discussions. The dataset reveals that rumour cascades exhibit faster diffusion dynamics, shorter half-lives, and a greater proportion of direct responses to the source tweet. These characteristics suggest that conversational behaviour may provide useful early indicators of misinformation before formal verification becomes available.

In contrast, NarraDetect provides a mechanism for measuring narrativity. The dataset contains human-grounded assessments of narrative structure and demonstrates clear separation between narrative and non-narrative texts. The strong agreement among narrativity dimensions further suggests that narrative characteristics can be represented as a coherent underlying construct.

A key difference between the datasets is document length. NarraDetect passages contain a median of approximately 104 –111 words, whereas PHEME tweets contain approximately 15 words. Consequently, narrativity cannot be assumed to transfer directly from long-form text to social media posts. Instead, NarraDetect is used to learn general narrative patterns that can subsequently be applied to Twitter discussions.

Taken together, the two datasets support different components of the proposed framework. PHEME contributes temporal, structural, and user-interaction signals, while NarraDetect contributes narrative signals. Integrating these sources enables the investigation of whether changes in narrativity and discussion structure can provide early indicators of emerging misinformation.

5. Implications for Methodology

The findings from the exploratory analysis directly inform the design of the proposed methodology.

First, the strong temporal differences observed in PHEME suggest that early engagement patterns, including reply arrival rates and thread half-life, should be incorporated as predictive features.

Second, differences in cascade composition indicate that conversational structure may be informative even when aggregate graph statistics such as depth and width are similar across classes.

Third, the limited separation provided by static textual features suggests that temporal evolution may be more informative than isolated content snapshots.

Finally, the NarraDetect analysis demonstrates that narrativity can be measured reliably using human-grounded annotations. This motivates the development of a narrativity classifier trained on NarraDetect and subsequently applied to PHEME discussions. The resulting narrativity scores can then be analysed alongside temporal and structural signals to investigate how narrative evolution unfolds during the spread of misinformation.